



Center for Advanced Studies in Engineering

Assignment no 1

Artificial Intelligence

Name: Kashaf Naveed

Regno: 2410-0058

Submitted to: Sir Hamid Ashfaq

Assignment No 01 "Artificial Intelligence"

Name: Kashaf Naveed

(BSCS - 5th)

Reg No: 2410-0058

Submitted to: Sir Hamid Ashfaq

"Water Jug Problem"

Problem Formulation

We have two jugs

- 4 liter jug
- 3 liter jug

Goal:- Measure exactly 2 liters in the 4-liter jug states:-

A state is represented as (x, y)

- x = amount of water in the 4-liter jug
- y = amount of water in the 3-liter jug

Example states:

$(0, 0) \rightarrow$ both jugs empty.

$(4, 0) \rightarrow$ 4 liter jug full.

$(1, 3) \rightarrow$ 1 liter in 4-liter jug and 3 liters in 3-liter jug.

Initial state: $(0, 0)$

- 1) Fill the 4-litre jug.
- 2) Fill the 3-liter jug.
- 3) Empty the 4-liter jug.
- 4) Empty the 3-liter jug.
- 5) Pour water from 4-liter jug $\rightarrow$ 3-liter jug.
- 6) Pour water from 3-liter jug $\rightarrow$ 4-liter jug.

Transition Model:-

- $(x, y) \rightarrow (4, y)$
- $(x, y) \rightarrow (x, 3)$
- $(x, y) \rightarrow (0, y)$
- $(x, y) \rightarrow (x, 0)$
- water moves until 3-liter jug is full
- water moves until 4-liter jug is full

Goal Test:-

Check if:-

$x=2$

Meaning 2-liters water in the 4-liter jug

Path cost:-

Each action cost $\Rightarrow$ 1 step

Total cost = number of steps taken

Algorithm (General)

- 1- Start from initial state $(0,0)$
- 2- Check if current state is goal state
- 3- Generate all possible next states using actions:
 - Fill jug
 - Empty jug
 - Pour water
- 4- Add new states to search structure (queue or stack)
- 5- Repeat until goal state is found.
- 6- Print the sequence of states.

Dry Run using BFS and DFS:

BFS:-

Start:-

$(0,0)$

Steps:-

$(0,0)$

$\rightarrow (4,0)$

$\rightarrow (1,3)$

$\rightarrow (1,0)$

$\rightarrow (0,1)$

$\rightarrow (4,1)$

$\rightarrow (2,3)$

$\rightarrow (2,0)$ Goal

DFS

Steps:-

$(0,0)$

$\rightarrow (0,3)$

$\rightarrow (3,0)$

$\rightarrow (3,3)$

$\rightarrow (4,2)$

$\rightarrow (0,2)$

$\rightarrow (2,0)$ Goal

Code

```
JUG1 = 4
JUG2 = 3
GOAL = 2

def water_jug_bfs():
    visited = set()
    queue = deque()

    queue.append((0, 0))
    visited.add((0, 0))

    while queue:
        x, y = queue.popleft()
        print("State:", (x, y))

        if x == GOAL:
            print("Goal reached! 4-liter jug has 2 liters.")
            return

        next_states = [
            (JUG1, y),
            (x, JUG2),
            (0, y),
            (x, 0),
```

```

        (0, y),
        (x, 0),
        (x - min(x, JUG2 - y), y + min(x, JUG2 - y)),
        (x + min(y, JUG1 - x), y - min(y, JUG1 - x))
    ]

    for state in next_states:
        if state not in visited:
            visited.add(state)
            queue.append(state)

water_jug_bfs()

```

Output

```

... State: (0, 0)
    State: (4, 0)
    State: (0, 3)
    State: (4, 3)
    State: (1, 3)
    State: (3, 0)
    State: (1, 0)
    State: (3, 3)
    State: (0, 1)
    State: (4, 2)
    State: (4, 1)
    State: (0, 2)
    State: (2, 3)
    Goal reached! 4-liter jug has 2 liters.

```

